

PA2 Report - Requirement 4: Use Case Specification

Group: 06

Course: CSC13112 - UI/UX Design

Assignment: Project Assignment 2

Product Scope: Freestyle Chess mobile website on smartphone browser

Document Purpose: Provide Use Case Diagrams (Mermaid flow models) and complete Use Case Specifications for each conceptual solution proposed in [06-PA2-ProjectProposal1.md](#).

1. Overview & Conceptual Solution Mapping

In [06-PA2-ProjectProposal1.md](#), Group 06 proposed six conceptual solutions across three user problems (**P-01: One-Handed Reachability**, **P-02: Lack of Sticky Search & Filtering**, and **P-05: Lack of Schedule Information**).

This document formalizes the actors, artifacts, interaction flows, and detailed use case specifications for each proposed solution.

1.1 Summary of Proposed Conceptual Solutions

Problem ID	Problem Description	Solution ID	Conceptual Solution Name	Target User Goal
P-01	Hamburger menu at top-left is out of reach for one-handed users	CS1.1	Persistent Bottom Navigation Bar	Direct 1-tap navigation to major sections within natural thumb sweep zone
P-01	Hamburger menu at top-left is out of reach for one-handed users	CS1.2	Draggable Floating Action Button (FAB)	Customizable, floating trigger that can be placed anywhere on screen
P-02	Lack of sticky search & filtering on long listing pages	CS2.1	Sticky Search Bar with Flexible Name Matching	Always-accessible text search accepting diacritics, partial names, and any word order
P-02	Lack of sticky search & filtering on long listing pages	CS2.2	Faceted Filter Chips (Tap-to-Filter)	Instant, zero-typing filtering by title (GM, IM), rating range, or event date
P-05	Schedule does not provide enough information, forcing users to look elsewhere	CS3.1	Additional Important Information Popup	Quick expandable preview of top performers or star players without layout overhaul
P-05	Schedule does not provide enough	CS3.2	Interactive Chess Schedule Calendar	Chronological calendar view with date navigation, search, and expanded event cards

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2. Actors and Artifacts Inventory

2.1 Actors

- **Mobile Visitor (Primary Actor):** General smartphone user browsing the Freestyle Chess mobile website in short sessions (2-3 minutes) while commuting, resting, or multi-tasking.
- **First-Time / Casual User:** Visitor unfamiliar with Freestyle Chess who needs quick orientation and simple, low-effort navigation.
- **Core Chess Follower:** Experienced chess enthusiast seeking fast access to specific player ratings, rankings, match developments, and detailed schedule information.

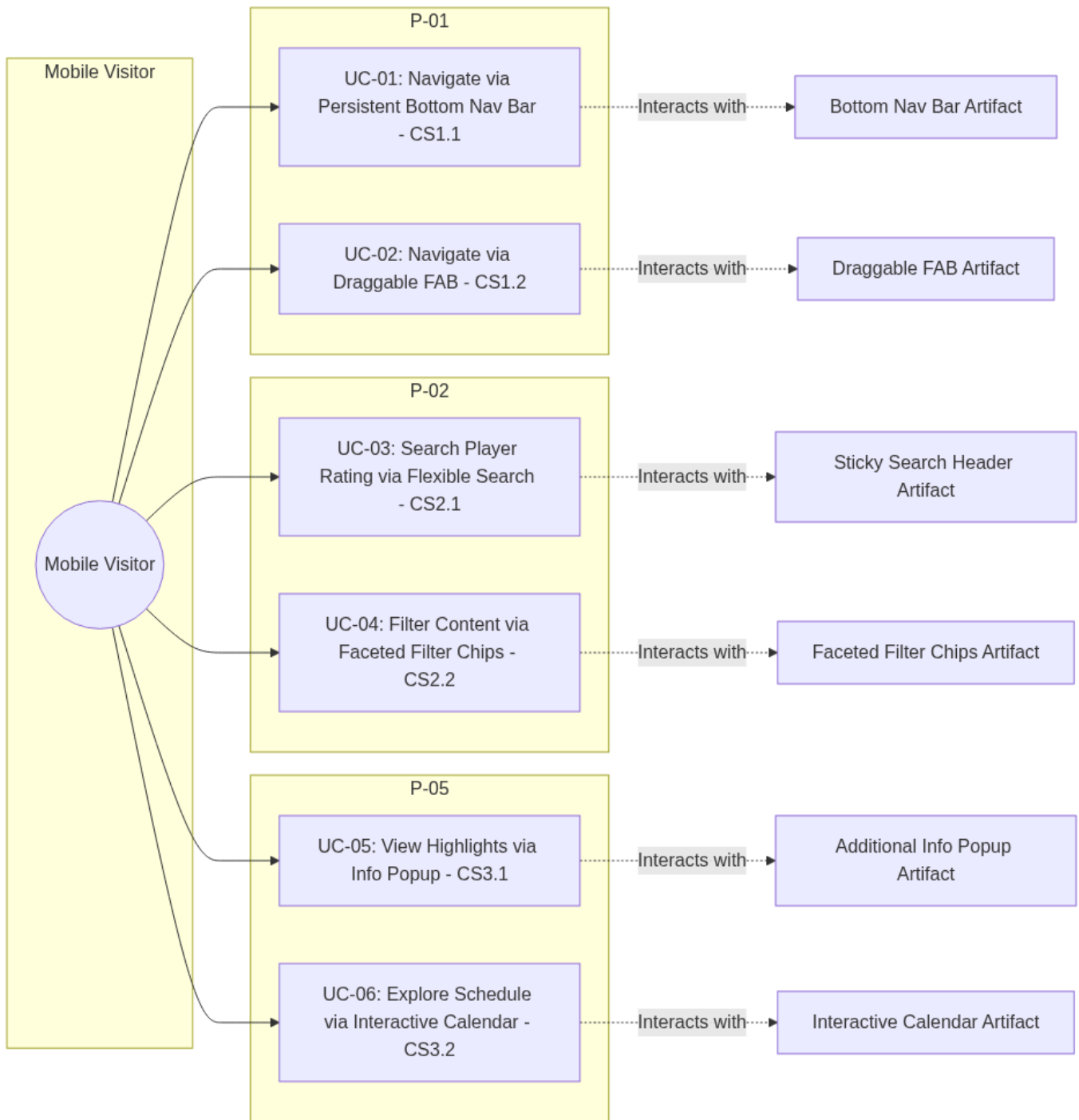
2.2 System & Interface Artifacts

- **Bottom Navigation Bar Artifact (ART-BNB):** Persistent 5-tab bar (Home, News, Schedule, Rating, Videos) fixed at the bottom viewport boundary ($y = \text{bottom}$).
- **Draggable FAB Artifact (ART-FAB):** Circular action button floating over page content, supporting press-and-drag gestures to re-anchor to bottom-right, bottom-left, or custom positions.
- **Sticky Search Header Artifact (ART-SSH):** Fixed header component at top viewport ($y = 0$) containing a text input field, clear button, and real-time auto-suggest dropdown.
- **Faceted Filter Chips Artifact (ART-FFC):** Horizontal scrollable row of interactive chip toggles (e.g., **GM**, **IM**, **Rating > 2700**, **Geller Cup 2026**) that apply instant client/server data filters.
- **Data Listing View Artifact (ART-DLV):** Dynamic list container (Rating Leaderboard or News Feed) that renders filtered results with skeleton loading indicators.
- **Additional Information Popup Artifact (ART-AIP):** Expandable overlay card rendered over schedule item rows, displaying top-3 results for past events or featured player names for upcoming events, with a direct "Read more" section link.
- **Interactive Schedule Calendar Artifact (ART-ISC):** Dedicated schedule page containing chronological match listings, vertical time swiping, a date-picker jump control, sticky search bar, and expandable event cards (ART-EEC).

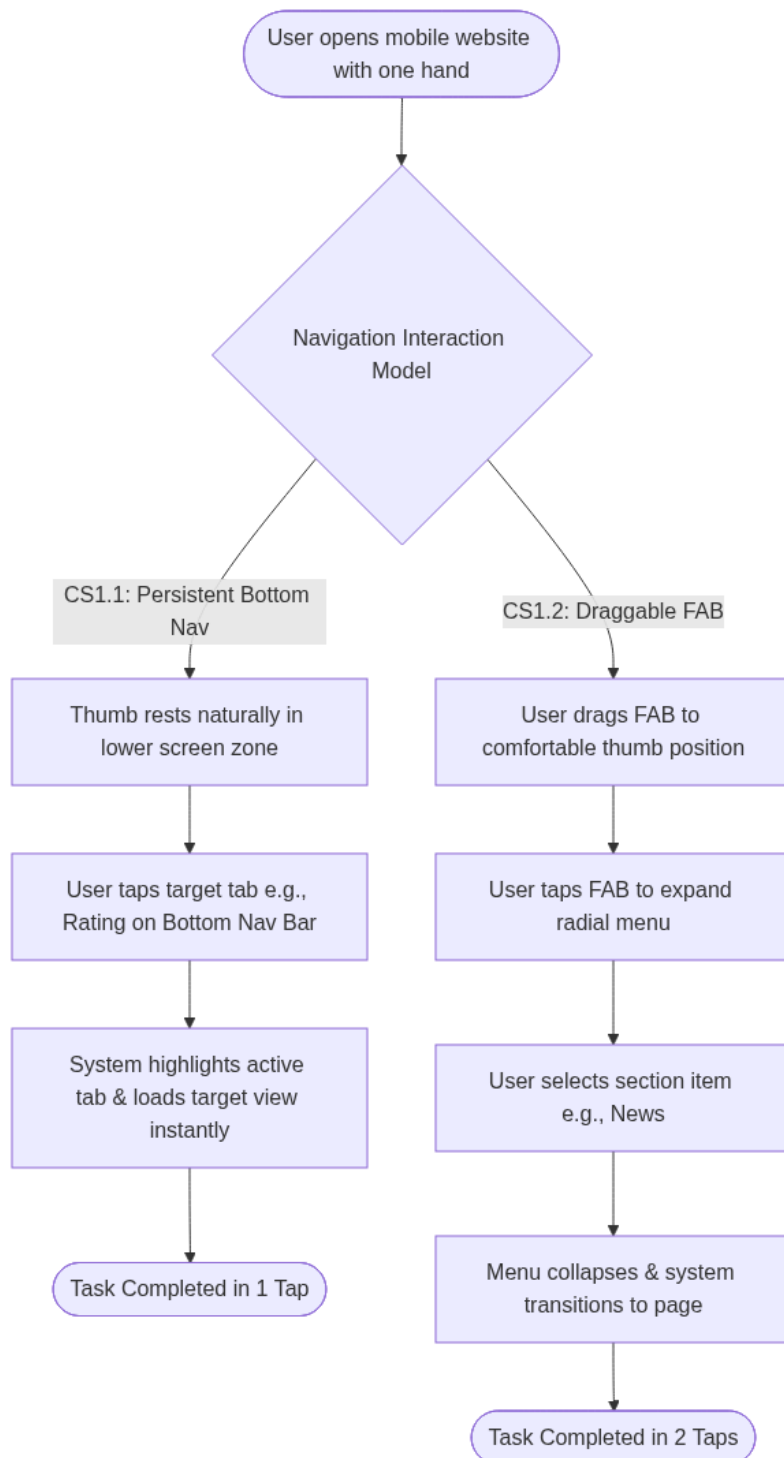
3. Use Case Diagrams & Flow Models

Below are the Mermaid Use Case diagrams and Interaction Flow models for each proposed conceptual solution.

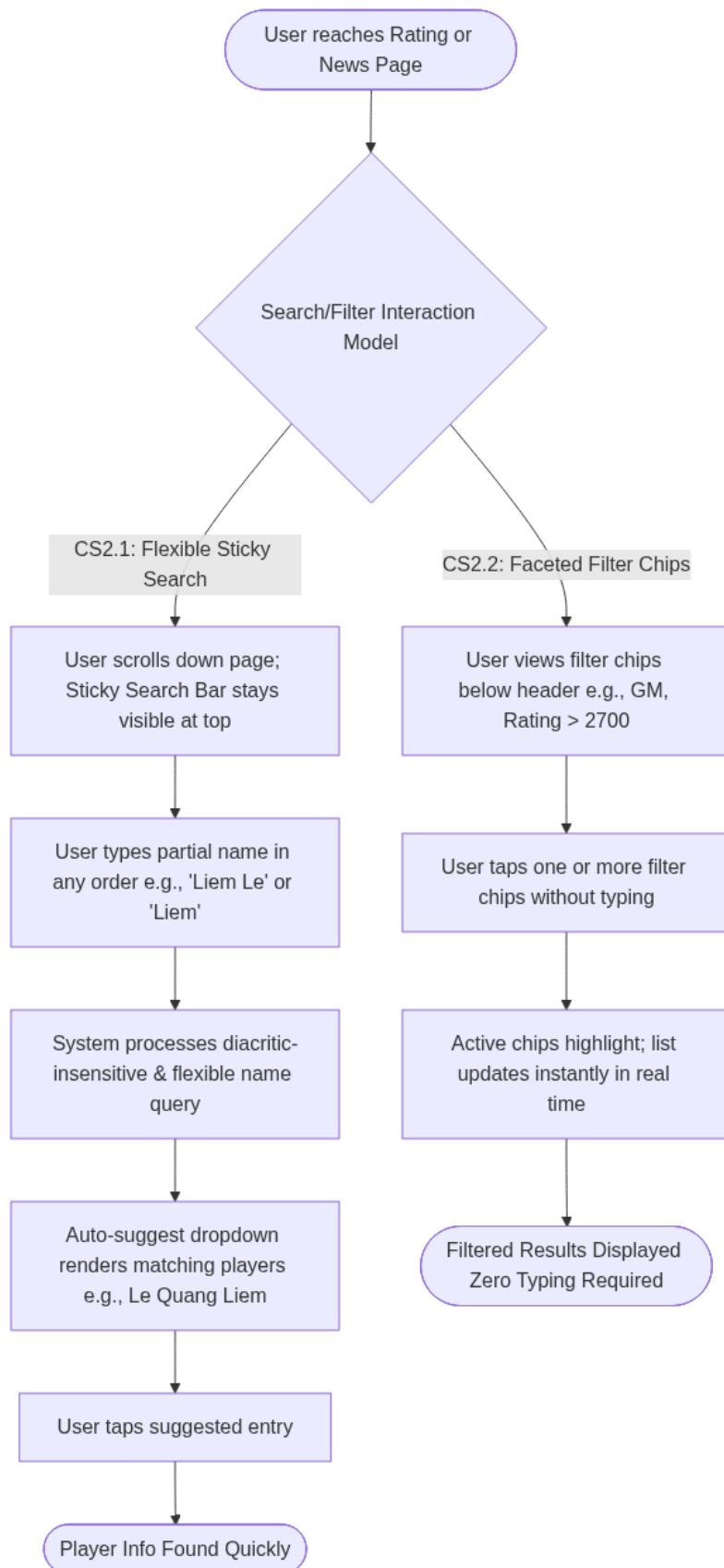
3.1 Use Case Diagram — Overall System Context



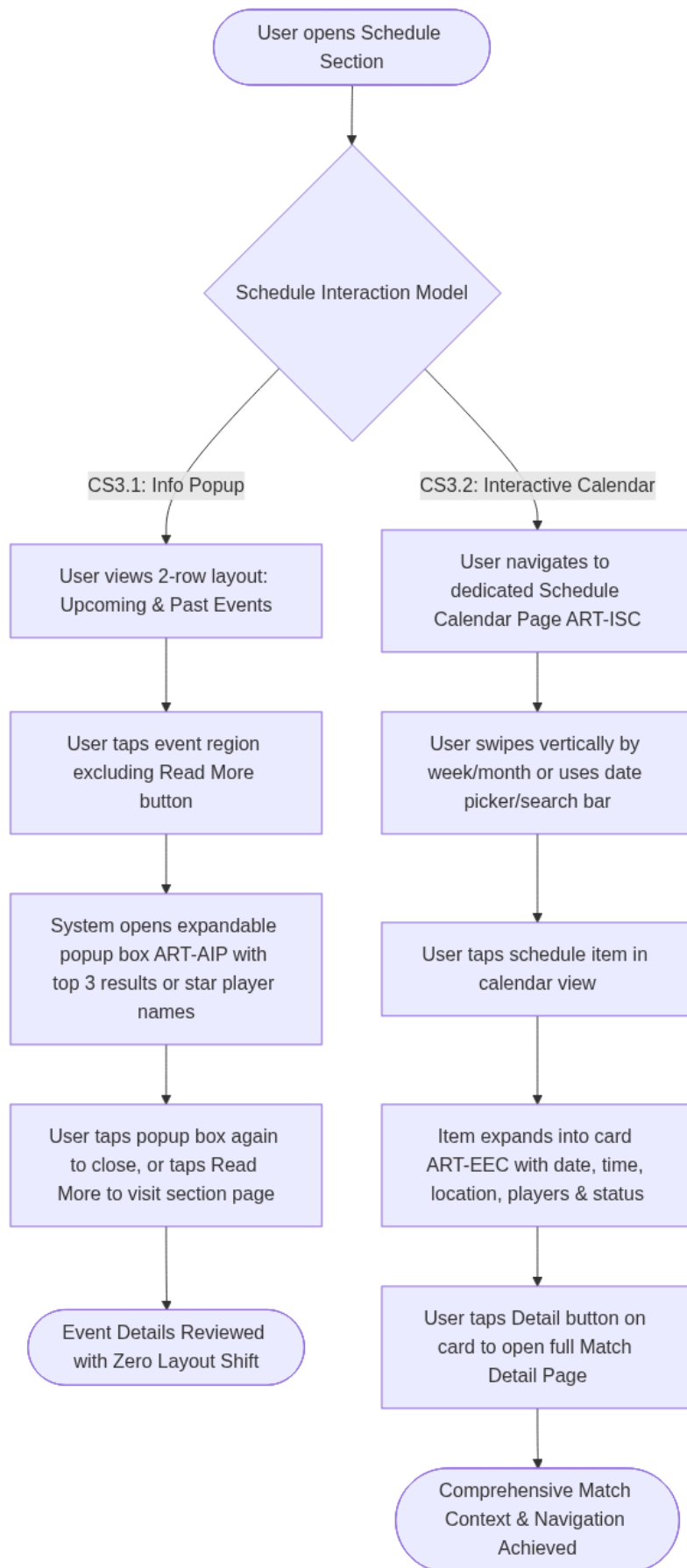
3.2 Flow Model — Conceptual Solution CS1.1 vs CS1.2 (Navigation)



3.3 Flow Model — Conceptual Solution CS2.1 vs CS2.2 (Search & Filtering)



3.4 Flow Model — Conceptual Solution CS3.1 vs CS3.2 (Schedule Exploration)



4. Use Case Specifications

4.1 UC-01: Navigate via Persistent Bottom Navigation Bar (CS1.1)

Field	Details
Use Case ID	UC-01
Use Case Name	Navigate to Main Sections via Persistent Bottom Navigation Bar
Related Concept	CS1.1 (Persistent Bottom Navigation Bar for P-01)
Primary Actor	Mobile Visitor (One-handed smartphone user)
Target Artifacts	Bottom Navigation Bar Artifact (ART-BNB), Main View Container (ART-DLV)
Preconditions	User is browsing any page of the Freestyle Chess mobile website on a smartphone browser.
Trigger	User wants to switch from the current page to another primary section (e.g., from Home to Rating).
Main Success Scenario	1. User Action: User looks at the lower portion of the screen where ART-BNB is fixed. 2. System Response: ART-BNB displays 5 distinct icons with text labels (<i>Home, News, Schedule, Rating, Videos</i>) within natural thumb reach. 3. User Action: User taps the "Rating" icon using their holding thumb without shifting grip. 4. System Response: System provides instant visual active state feedback (icon highlight), triggers a page loading indicator, and renders the Rating Leaderboard view. 5. User Action: User begins scanning the Rating page content.
Alternative Flows	ALT-1 (Repeated tap on active tab): If user taps the icon of the page they are currently on, system smooth-scrolls the view back to the top of the page.
Exception Flows	EX-1 (Network latency): If page content fails to load within 2 seconds, system displays a skeleton loader inside ART-DLV and a "Retry" prompt while keeping ART-BNB active.
Postconditions	User successfully arrives at the selected section with zero grip adjustments.
Empirical Evidence	Validates N03 ; resolves reachability complaint from 43.1% of survey users (Q13) and direct interview struggles from P01, P02, P03.

4.2 UC-02: Access Navigation Menu via Draggable Floating Action Button (CS1.2)

Field	Details
Use Case ID	UC-02
Use Case Name	Access Navigation Menu via Draggable Floating Action Button (FAB)
Related Concept	CS1.2 (Draggable FAB for P-01)
Primary Actor	Mobile Visitor (Left-handed, right-handed, or situational grip user)
Target Artifacts	Draggable FAB Artifact (ART-FAB), Floating Navigation Drawer (ART-FND)

Field	Details
Preconditions	User is on the mobile website; ART-FAB is visible near the bottom-right corner.
Trigger	User wants to access navigation while holding the phone in a non-standard or left-handed grip.
Main Success Scenario	1. User Action: User presses and holds ART-FAB, then drags it to a comfortable thumb position (e.g., bottom-left corner). 2. System Response: System highlights ART-FAB with drop-shadow feedback and smoothly repositions the button following touch coordinates. 3. User Action: User releases touch; system snaps ART-FAB to the nearest edge and saves position to local storage. 4. User Action: User taps ART-FAB. 5. System Response: ART-FAB transforms into ART-FND (radial menu or bottom sheet) displaying section shortcuts. 6. User Action: User taps "Schedule". 7. System Response: System navigates to Schedule view and collapses ART-FND back into ART-FAB.
Alternative Flows	ALT-1 (Open without dragging): If the user taps ART-FAB without a hold-and-drag gesture, the system opens ART-FND directly.
Exception Flows	EX-1 (Accidental Drag): If drag distance is under 5px, system treats touch as a tap and opens ART-FND .
Postconditions	Navigation trigger is custom-anchored to the user's exact physical thumb reach.
Empirical Evidence	Validates N04 ; directly answers P05's request for a customizable draggable control and supports 41.7% of users with situational grips (Q4).

4.3 UC-03: Search Player Rating via Flexible Sticky Search Bar (CS2.1)

Field	Details
Use Case ID	UC-03
Use Case Name	Search Player Rating with Flexible Name Order & Auto-Suggestions
Related Concept	CS2.1 (Sticky Search Bar with Flexible Name Matching for P-02)
Primary Actor	Chess Follower / Mobile Visitor
Target Artifacts	Sticky Search Header Artifact (ART-SSH), Data Listing View Artifact (ART-DLV)
Preconditions	User is on the Rating Leaderboard page containing over 100 player entries.
Trigger	User wants to find rating information for a specific player (e.g., "Le Quang Liem") without scrolling.
Main Success Scenario	1. User Action: User scrolls down the leaderboard; ART-SSH remains fixed at the top viewport ($y = 0$). 2. User Action: User taps the search text field in ART-SSH.

Field	Details
	3. System Response: Virtual keyboard appears; ART-SSH maintains top position. 4. User Action: User types partial name in natural order (e.g., "Liem Le" or "liem"). 5. System Response: System executes diacritic-insensitive, multi-word token query across database fields (FirstName, LastName, FullName). 6. System Response: ART-SSH renders auto-suggest list showing matching player cards (e.g., "GM Le Quang Liem - Rating 2736"). 7. User Action: User taps the suggested player card. 8. System Response: System highlights the selected player row and scrolls smoothly to target entry.
Alternative Flows	ALT-1 (No exact match): If query yields no matches (e.g., "Liemmm"), system displays "No player found" with suggestions ("Did you mean: Liem Le?"). ALT-2 (Clear search): User taps "X" clear button in ART-SSH to restore full leaderboard.
Exception Flows	EX-1 (Offline / Connection drop): System relies on cached client-side index to serve auto-suggestions without freezing UI.
Postconditions	User locates target player rating within 3-5 seconds without excessive vertical scrolling.
Empirical Evidence	Validates N11 , N23 ; directly fixes failed searches observed in P06 and P09, and supports 61.1% of users struggling with exact memory recall (Q19).

4.4 UC-04: Filter Content via Faceted Filter Chips (CS2.2)

Field	Details
Use Case ID	UC-04
Use Case Name	Filter Content via Faceted Filter Chips (Zero-Typing Mode)
Related Concept	CS2.2 (Faceted Filter Chips for P-02)
Primary Actor	First-Time Visitor / Event-Driven Viewer
Target Artifacts	Faceted Filter Chips Artifact (ART-FFC), Data Listing View Artifact (ART-DLV)
Preconditions	User is browsing the Rating Leaderboard or Tournament News Feed.
Trigger	User wants to view specific subsets of information (e.g., Grandmasters only, or Geller Cup news) without typing.
Main Success Scenario	1. User Action: User views horizontal row of ART-FFC positioned directly below page title. 2. System Response: ART-FFC presents clear category chips (All, GM, IM, Rating > 2700, 2026 Events). 3. User Action: User taps the "GM" (Grandmaster) chip. 4. System Response: System toggles chip visual state to Active (filled background), filters list data instantly, and updates ART-DLV.

Field	Details
	5. User Action: User taps a secondary filter chip ("Rating > 2700"). 6. System Response: System applies combined filter query (Title == GM AND Rating > 2700) and displays refined list of 12 top players. 7. User Action: User reviews filtered results.
Alternative Flows	ALT-1 (Deselect filter): User taps active "GM" chip again to deselect it; system updates view to show all titles.
Exception Flows	EX-1 (Empty result set): If active chip combination returns 0 items, system displays clear empty state ("No players match these filters") with a 1-tap "Reset Filters" button.
Postconditions	User discovers specific target information through simple single-tap interactions without needing keyboard entry.
Empirical Evidence	Validates N12, N27 ; directly answers Q17 preference (chosen by 31 users / 81.9% wanting search/filter support) and resolves P10's inability to search by title or rating range.

4.5 UC-05: View Event Highlights via Additional Information Popup (CS3.1)

Field	Details
Use Case ID	UC-05
Use Case Name	View Event Highlights via Additional Information Popup
Related Concept	CS3.1 (Additional Important Information Popup for P-05)
Primary Actor	Core Chess Follower / Mobile Visitor
Target Artifacts	Additional Information Popup Artifact (ART-AIP), Schedule Row Item (ART-SRI)
Preconditions	User is viewing the Schedule section retaining the familiar 2-row layout (<i>Upcoming Events</i> and <i>Past Events</i>).
Trigger	User wants to quickly see key match context (such as featured star grandmasters or top-3 winners) without navigating away from the page.
Main Success Scenario	1. User Action: User scans event items in either the Upcoming or Past events row. 2. User Action: User touches an event item region (outside the explicit "Read more" link). 3. System Response: System highlights touch region and displays expandable overlay popup (ART-AIP) over the event card with smooth drop-shadow animation. 4. System Response: ART-AIP displays curated highlights: top-3 players/results for past matches, or key big-name participants for upcoming events. 5. User Action: User reviews information and taps anywhere inside ART-AIP again. 6. System Response: System collapses ART-AIP smoothly, restoring normal row appearance.

Field	Details
Alternative Flows	ALT-1 (Direct Section Navigation): User touches "Read more" directly inside ART-AIP or on the event row; system navigates to the dedicated section page. ALT-2 (Switching Active Popups): User touches a different event row while ART-AIP is open; system closes the previous popup and expands the new event popup instantly.
Exception Flows	EX-1 (Missing Highlight Data): If player/result highlight data is unavailable, ART-AIP displays fallback message ("Participant list updating shortly") alongside the "Read more" link.
Postconditions	User obtains key player names or match results in 1 tap without losing their position on the schedule page.
Empirical Evidence	Validates N24 , N28 ; addresses P09 's insufficient-match-detail feedback while preserving layout familiarity for returning users. Evidence for the underlying problem (P-05) comes from a single dedicated participant rather than broad survey data; see 06-PA2-UserAnalysis.md scoring rationale for why the group carried it forward regardless.

4.6 UC-06: Explore Match Schedule via Interactive Chess Calendar (CS3.2)

Field	Details
Use Case ID	UC-06
Use Case Name	Explore Match Schedule via Interactive Chess Calendar
Related Concept	CS3.2 (Interactive Chess Schedule Calendar for P-05)
Primary Actor	Core Chess Follower / Mobile Visitor
Target Artifacts	Interactive Schedule Calendar Artifact (ART-ISC), Expanded Event Card (ART-EEC), Match Detail Page (ART-MDP)
Preconditions	User selects the Schedule tab from primary navigation; system opens dedicated ART-ISC page.
Trigger	User wants to explore past/upcoming chess matches chronologically, jump to a specific date, or search for a player's schedule.
Main Success Scenario	1. User Action: User lands on ART-ISC page showing matches organized chronologically by date and start time. 2. System Response: System visually highlights current date marker; displays directional arrow if current date is outside visible viewport. 3. User Action: User swipes vertically (downward for upcoming weeks, upward for past weeks) or selects a specific date via date picker. 4. System Response: System smoothly scrolls and renders calendar items for the selected timeframe. 5. User Action: User taps a schedule item.

Field	Details
	6. System Response: Item expands in-place into ART-EEC showing time, location, players, tournament name, match status, short description, and a "Detail" button. 7. User Action: User taps "Detail" button. 8. System Response: System navigates to ART-MDP (Match Detail Page) with complete player statistics and game history.
Alternative Flows	ALT-1 (Schedule Search): User types player name, tournament, or location into search bar; system filters calendar entries dynamically. ALT-2 (Collapse Card): User touches ART-EEC again or taps outside it; card collapses back into single-line schedule item. ALT-3 (Preserve State on Return): User returns from ART-MDP via back button; system preserves scroll position, selected date, search query, and expanded card.
Exception Flows	EX-1 (High Event Density): If many matches occur on one day, system displays top 3 events with a "View more" expander to prevent visual overcrowding.
Postconditions	User gains complete chronological orientation, flexible date jump & search, and multi-level information (summary -> expanded card -> detail page).
Empirical Evidence	Validates N20, N21, N24, N28, N29 ; directly resolves P09 's feedback by supplying full match info (time, players, location, format) and replaces misleading chevron icon with a dedicated calendar mental model. Evidence for the underlying problem (P-05) comes from a single dedicated participant rather than broad survey data; see 06-PA2-UserAnalysis.md scoring rationale for why the group carried it forward regardless.

5. Summary & Verification Matrix

The six Use Case Specifications above cover 100% of the conceptual solutions proposed in **06-PA2-ProjectProposal.md**.

Use Case ID	Conceptual Solution	Problem Addressed	User Interaction Type	Verification Method
UC-01	CS1.1: Persistent Bottom Nav	P-01: Reachability	1-tap direct navigation	Task completion time < 2s; zero thumb stretch
UC-02	CS1.2: Draggable FAB	P-01: Reachability	Touch-drag & 2-tap radial menu	Custom position persistence across sessions
UC-03	CS2.1: Flexible Sticky Search	P-02: Long List Search	Text input + auto-suggest	Success rate on reversed name order queries
UC-04	CS2.2: Faceted Filter Chips	P-02: Long List Search	1-tap filter toggles	Zero-typing filtering by title and event date
UC-05	CS3.1: Additional Info Popup	P-05: Lack of Schedule Info	1-tap expandable overlay box	Instant highlight access without layout disruption

Use Case ID	Conceptual Solution	Problem Addressed	User Interaction Type	Verification Method
UC-06	CS3.2: Interactive Calendar	P-05: Lack of Schedule Info	Calendar swipe, date jump & expanded card	Full temporal navigation & 3-stage progressive disclosure